

HDOC 21

Name: Keshav Mittal

SAP ID: 590032934

Batch: 16

1. Algorithm

1. Start
2. Read two integers m and n.
3. Validate Inputs: If $m \leq 0$, $n \leq 0$, or $m > n$, display an error message and repeat step 2 until valid input is provided.
4. Display the menu with options 1 to 5.
5. Read user's choice.
6. Use a switch statement based on choice:
 - Case 1 (Multiples): Loop i from m to n. Inside, loop j from 1 to 10. Print $i * j$.
 - Case 2 (Addition): Loop i from m to n. Inside, loop j from 1 to 10. Print $i + j$.
 - Case 3 (Subtraction): Loop i from m to n. Inside, loop j from 1 to 10. Print $i - j$.
 - Case 4 (Division): Loop i from m to n. Inside, loop j from 1 to 10. Print $(\text{float})i / j$ formatted to two decimal places.
 - Case 5 (Exit): Display termination message and break.
 - Default: Display "Invalid choice" error message.
7. Repeat steps 4-6 until choice == 5.
8. Stop

2. Test-Case Table

TC ID	Scenario Description	Inputs (m, n, choice)	Expected Output
TC-01	Invalid range input	m: 5, n: 2	Error: Ensure m and n are positive and $m \leq n$.
TC-02	Multiples table (Valid)	m: 2, n: 3, choice: 1	For 2: 2 4 6 8 10 12 14 16 18 20 For 3: 3 6 9 12 15 18 21 24 27 30
TC-03	Addition table (Valid)	m: 4, n: 4, choice: 2	For 4: 5 6 7 8 9 10 11 12 13 14
TC-04	Subtraction table (Valid)	m: 5, n: 5, choice: 3	For 5: 4 3 2 1 0 -1 -2 -3 -4 -5
TC-05	Division table (Valid)	m: 1, n: 2, choice: 4	For 1: 1.00 0.50 0.33 0.25 ... For 2: 2.00 1.00 0.67 0.50 ...
TC-06	Invalid menu choice	choice: 8	Invalid choice. Please enter 1-5.
TC-07	Exit program	choice: 5	Exiting program... (Program terminates)

3. C Program Source Code

```
#include<stdio.h> int main() {
    int m,n, choice; int i, j;

    // Input m and n with validation do {
        printf("Enter two positive integers m and n (m <= n): "); scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n) {
            printf("Error: Ensure m and n are positive and m <= n.\n");
        }
    } while (m <= 0 || n <= 0 || m > n);

    // Menu driven loop do {
        printf("\n--- Table Operations Menu ---\n"); printf("1. Print first 10 multiples\n"); printf("2. Print addition table\n"); printf("3. Print subtraction table\n"); printf("4. Print division table\n"); printf("5. Exit\n");
        printf("Enter your choice: "); scanf("%d", &choice);

        switch (choice) { case 1:
            printf("\n--- Multiples ---\n"); for (i = m; i <= n; i++) {
                printf("Multiples of %d:\t", i); for (j = 1; j <= 10; j++) {
                    printf("%d ", i * j);
                }
                printf("\n");
            }
            break;

        case 2:
            printf("\n--- Addition Table ---\n"); for (i = m; i <= n; i++) {
                printf("Addition for %d:\t", i); for (j = 1; j <= 10; j++) {
                    printf("%d ", i + j);
                }
                printf("\n");
            }
            break;

        case 3:
            printf("\n--- Subtraction Table ---\n"); for (i = m; i <= n; i++)
            {
                printf("Subtraction for %d:\t", i); for (j = 1; j <= 10; j++) {
```

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        printf("%d ", i - j);
    }
    printf("\n");
}
break;

case 4:
    printf("\n--- Division Table ---\n");
    for (i = m; i <= n; i++) {
        printf("Division for %d:\t", i);
        for (j = 1; j <= 10; j++) {
            printf("%.2f ", (float)i / j);
        }
        printf("\n");
    }
    break;

case 5:
    printf("Exiting program...\n");
    break;

default:
    printf("Error: Invalid choice. Please enter 1-5.\n");
}
} while (choice != 5);

return 0;
}

```